

CBSE Class 12 Chemistry — Half-Yearly Predicted Paper (Set 8 of 10)

50 Most-Probable PYQs & Board-Pattern Practice Questions

Section A: 1-Mark Questions (MCQs / Assertion-Reason)

Q1. A mixture of ethanol and water boils at a temperature lower than the boiling points of both pure ethanol and pure water. This indicates that the solution: (a) Shows positive deviation from Raoult's law (b) Shows negative deviation from Raoult's law (c) Forms a maximum boiling azeotrope (d) Behaves as an ideal solution [Solutions] [PYQ 2019] [Confidence: Very High]

Q2. Which of the following ions is coloured in an aqueous solution? (a) Sc^{3+} (b) Ti^{4+} (c) V^{3+} (d) Zn^{2+} [The d- and f-Block Elements] [Practice] [Confidence: High]

Q3. In the Arrhenius equation, $k = Ae^{-E_a/RT}$, the pre-exponential factor (A) corresponds to: (a) The fraction of molecules possessing energy greater than E_a (b) The activation energy of the reaction (c) The collision frequency of the reacting molecules (d) The rate of the reaction at absolute zero [Chemical Kinetics] [PYQ 2017] [Confidence: Very High]

Q4. During the reaction of aldehydes with semicarbazide ($H_2N - NH - CO - NH_2$) to form semicarbazones, which specific nitrogen atom acts as the nucleophile? (a) The nitrogen atom bonded directly to the carbonyl group (b) The nitrogen atom of the NH_2 group farthest from the carbonyl group (hydrazine part) (c) The nitrogen atom in the NH group (d) All three nitrogen atoms attack equally [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2015] [Confidence: High]

Q5. According to Kohlrausch's law, the limiting molar conductivity of NH_4OH , $\Lambda_m^\circ(NH_4OH)$, is exactly equal to: (a) $\Lambda_m^\circ(NH_4Cl) + \Lambda_m^\circ(NaOH) - \Lambda_m^\circ(NaCl)$ (b) $\Lambda_m^\circ(NH_4Cl) + \Lambda_m^\circ(NaCl) - \Lambda_m^\circ(NaOH)$ (c) $\Lambda_m^\circ(NH_4Cl) + \Lambda_m^\circ(NaOH) + \Lambda_m^\circ(NaCl)$ (d) $\Lambda_m^\circ(NaOH) + \Lambda_m^\circ(NaCl) - \Lambda_m^\circ(NH_4Cl)$ [Electrochemistry] [PYQ 2018] [Confidence: Very High]

Q6. Assuming complete dissociation, which of the following 0.1 M aqueous solutions will have the highest freezing point? (a) $NaCl$ (b) $CaCl_2$ (c) $Al_2(SO_4)_3$ (d) $Urea$ [Solutions] [Practice] [Confidence: High]

Q7. The most common and stable oxidation state exhibited by the lanthanoid series elements is: (a) +2 (b) +3 (c) +4 (d) +6 [The d- and f-Block Elements] [PYQ 2020] [Confidence: Very High]

Q8. The addition of one equivalent of a monohydric alcohol to an aldehyde in the presence of dry HCl gas yields: (a) An acetal (b) A hemiacetal (c) A ketal (d) A carboxylic acid [Aldehydes, Ketones and Carboxylic Acids] [PYQ SQP 2022] [Confidence: Very High]

Q9. Why is chloroform ($CHCl_3$) strictly stored in closed dark-coloured bottles completely filled up to the brim? (a) It is highly volatile and boils at room temperature. (b) It slowly oxidizes in air and sunlight to form the highly poisonous gas phosgene. (c) It undergoes spontaneous disproportionation. (d) It reacts with the silica in glass in the presence of light. [Haloalkanes and Haloarenes] [PYQ 2016] [Confidence: Very High]

Q10. The rate of a chemical reaction is given by $Rate = k[A]^{1/2}[B]^{1/2}$. The correct unit of the specific rate constant (k) for this reaction is: (a) $mol\ L^{-1}\ s^{-1}$ (b) s^{-1} (c) $L\ mol^{-1}\ s^{-1}$ (d) $mol^{1/2}\ L^{-1/2}\ s^{-1}$ [Chemical Kinetics] [Practice] [Confidence: Medium]

Q11. The IUPAC name of the ether $\text{CH}_3 - \text{O} - \text{CH}(\text{CH}_3)_2$ is: (a) Isopropyl methyl ether (b) 1-Methoxypropane (c) 2-Methoxypropane (d) Methoxyisopropane [Alcohols, Phenols and Ethers] [PYQ 2014] [Confidence: Very High]

Q12. Which of the following compounds will NOT undergo the Cannizzaro reaction? (a) Formaldehyde (HCHO) (b) Trichloroacetaldehyde (CCl_3CHO) (c) Benzaldehyde ($\text{C}_6\text{H}_5\text{CHO}$) (d) Acetaldehyde (CH_3CHO) [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2019] [Confidence: Very High]

Directions for Q13 to Q16: A statement of Assertion (A) followed by a statement of Reason (R) is given. Choose the correct answer out of the following choices: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

Q13. Assertion (A): An aqueous solution of copper sulphate can be safely stored in a zinc vessel. Reason (R): Zinc is situated above copper in the electrochemical series and is more reactive than copper. [Electrochemistry] [PYQ 2015] [Confidence: Very High]

Q14. Assertion (A): The coordination complex $[\text{Ni}(\text{CO})_4]$ is entirely diamagnetic in nature. Reason (R): Carbon monoxide (CO) is a strong field ligand and forces the pairing of $4s$ electrons of nickel into the $3d$ orbitals. [Coordination Compounds] [PYQ 2018] [Confidence: Very High]

Q15. Assertion (A): Benzyl chloride readily undergoes nucleophilic substitution via the $\text{S}_{\text{N}}1$ mechanism. Reason (R): The benzyl carbocation formed as an intermediate is highly stabilized by resonance. [Haloalkanes and Haloarenes] [PYQ 2020] [Confidence: Very High]

Q16. Assertion (A): Ortho-nitrophenol is less soluble in water than para-nitrophenol. Reason (R): Ortho-nitrophenol exhibits strong intramolecular hydrogen bonding, preventing it from forming extensive hydrogen bonds with water. [Alcohols, Phenols and Ethers] [PYQ 2017] [Confidence: High]

Section B: 2-Mark Questions (Short Answer)

Q17. Define the term *Ebullioscopic Constant* (K_b). Write its standard SI unit. [Solutions] [PYQ 2014] [Confidence: Very High]

Q18. While determining the resistance of an electrolytic solution in a conductivity cell, why is an alternating current (AC) source strictly used instead of a direct current (DC) source? [Electrochemistry] [PYQ 2019] [Confidence: Very High]

Q19. What is meant by a *pseudo-first-order reaction*? Give one specific chemical example with its chemical equation to illustrate it. [Chemical Kinetics] [PYQ 2015] [Confidence: High]

Q20. Compare the chemistry of Lanthanoids and Actinoids. Give any two specific points of difference in their chemical behaviour. [The d- and f-Block Elements] [PYQ 2016] [Confidence: Very High]

Q21. Using strict IUPAC norms, write the exact formulae for the following coordination compounds: (a) Potassium trioxalatoaluminate(III) (b) Dichloridobis(ethane-1,2-diamine)cobalt(III) ion [Coordination Compounds] [PYQ 2018] [Confidence: Very High]

Section C: 3-Mark Questions (Short Answer)

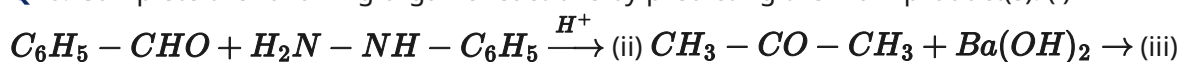
Q22. A solution is prepared by dissolving **30 g** of a non-volatile solute in **90 g** of water. It has a vapour pressure of **2.8 kPa** at **298 K**. Further, **18 g** of water is then added to the solution, and the new vapour pressure becomes **2.9 kPa** at **298 K**. Calculate the molar mass of the unknown solute. [Solutions] [PYQ 2017] [Confidence: Very High]

Q23. The electrical resistance of a conductivity cell containing a **0.001 M KCl** solution at **298 K** is **1500 Ω** . What is the cell constant (**G^***) of the cell if the conductivity of the **0.001 M KCl** solution at **298 K** is precisely **$0.146 \times 10^{-3} \text{ S cm}^{-1}$** ? [Electrochemistry] [PYQ 2018] [Confidence: Very High]

Q24. Give brief and specific chemical reasons for the following observations: (a) Chloroform is always stored in closed, dark-coloured bottles. (b) Iodoform gives a yellow precipitate with aqueous **AgNO_3** on heating, but chloroform does not give any precipitate. (c) Haloarenes are extremely less reactive towards nucleophilic substitution reactions than haloalkanes. [Haloalkanes and Haloarenes] [PYQ 2015 / 2020] [Confidence: Very High]

Q25. Write the exact chemical equations along with necessary reagents to carry out the following organic conversions: (a) Propene to Propan-2-ol (b) Benzyl chloride to Benzyl alcohol (c) Anisole to p-Bromoanisole [Alcohols, Phenols and Ethers] [PYQ 2014 / 2016] [Confidence: High]

Q26. Complete the following organic reactions by predicting the main product(s): (i)



$\text{CH}_3 - \text{CH}_2 - \text{COOH} + \text{Cl}_2 \xrightarrow{\text{Red P}_4}$ [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2019 / Practice] [Confidence: Very High]

Q27. Account for the following observations in d-block chemistry: (a) Transition metals are highly capable of forming complex compounds. (b) The standard electrode potential **E°** for the **$\text{Mn}^{3+} / \text{Mn}^{2+}$** couple is highly positive. (c) Zinc (**$Z = 30$**) is not considered a true transition metal. [The d- and f-Block Elements] [PYQ 2018 / 2023] [Confidence: Very High]

Q28. The half-life for the radioactive decay of Carbon-14 is exactly **5730 years**. An archaeological artifact containing wood was tested and had only **80%** of the Carbon-14 activity found in a living tree. Estimate the approximate age of the sample. (Given: **$\log(1.25) = 0.0969$**). [Chemical Kinetics] [PYQ 2016] [Confidence: High]

Section D: 4-Mark Questions (Case-Study)

Q29. Case Study: Isomerism in Coordination Compounds Coordination compounds exhibit various types of isomerism, which drastically affect their physical and chemical properties. Structural isomerism includes linkage, coordination, ionization, and solvate isomerism. Linkage isomerism arises when an ambidentate ligand is present. Ionization isomerism occurs when a counter ion displaces a ligand inside the coordination sphere. Furthermore, stereoisomerism includes geometrical (cis-trans) and optical isomerism, typically seen in square planar and octahedral complexes depending on their symmetry. (a) What specific type of structural isomerism is shown by the pair of complexes **$[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Br}$** and **$[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$** ? (1 Mark) (b) Give one example of a symmetrical bidentate ligand that commonly leads to optical isomerism when forming octahedral complexes (e.g., in a **$[\text{M}(\text{L} - \text{L})_3]$** system). (1 Mark) (c) Draw the structures of the *cis* and *trans* isomers of the square planar complex **$[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$** . Identify which of these two isomers possesses a net zero dipole moment. (2 Marks) [Coordination Compounds] [CBSE Term-2 2022 / Practice] [Confidence: Very High]

Q30. Case Study: Acidity of Carboxylic Acids Carboxylic acids are amongst the most acidic organic compounds, though they are much weaker than mineral acids. Their acidity is heavily dependent on the stability of the resulting carboxylate anion formed after losing a proton. The presence of electron-withdrawing groups (EWGs) on the alkyl or aryl chain stabilizes the carboxylate anion through inductive (-I) or resonance (-R) effects, thereby increasing the acid strength. Conversely, electron-donating groups (EDGs) destabilize the anion and decrease acidity. (a) Why is trifluoroacetic acid (**CF_3COOH**) a significantly stronger acid than trichloroacetic acid (**CCl_3COOH**)? (1 Mark) (b) What is the general effect

of an electron-donating group (like $-\text{CH}_3$ or $-\text{OCH}_3$) on the acidity of benzoic acid? (1 Mark) (c) Arrange the following compounds in the strictly increasing order of their acid strength: p-nitrobenzoic acid, benzoic acid, p-toluic acid (p-methylbenzoic acid). Briefly justify your arrangement. (2 Marks) [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2020 / Practice] [Confidence: Very High]

Section E: 5-Mark Questions (Long Answer)

Q31. (a) State Kohlrausch's law of independent migration of ions. State any one important application of this law in electrochemistry. (2 Marks) (b) Calculate the electromotive force (EMF) of the following galvanic cell at **298 K**: $\text{Pt}(s)|\text{Br}_2(l)|\text{Br}^-(0.010\text{ M})||\text{H}^+(0.030\text{ M})|\text{H}_2(g, 1\text{ bar})|\text{Pt}(s)$ Given: The standard reduction potential $E^\circ_{\text{Br}_2/\text{Br}^-} = +1.09\text{ V}$. (Assume $\frac{2.303RT}{F} = 0.0591\text{ V}$). (3 Marks) [Electrochemistry] [PYQ 2017 / Practice] [Confidence: Very High]

Q32. An organic compound **A** having the molecular formula $\text{C}_6\text{H}_6\text{O}$ gives a characteristic violet colour with an aqueous neutral FeCl_3 solution. When **A** is treated with CO_2 and NaOH at **400 K** under pressure, it yields compound **B**, which upon acidification gives compound **C**. Compound **C** readily reacts with acetyl chloride (CH_3COCl) in the presence of an acid catalyst to produce compound **D**, which is a widely used painkiller and antipyretic medicine. (a) Deduce the exact structures and standard chemical names of compounds **A**, **B**, **C**, and **D**. (4 Marks) (b) What is the specific named reaction used to convert compound **A** into compound **C**? (1 Mark) [Alcohols, Phenols and Ethers] [PYQ 2019 / 2023] [Confidence: Very High]

Q33. (a) The rate of a first-order reaction is dependent on the concentration of the reactant. Derive the integrated rate equation for a generic first-order reaction: $\text{R} \rightarrow \text{P}$. (3 Marks) (b) The specific rate constant for the first-order decomposition of Dinitrogen pentoxide (N_2O_5) is exactly $5.0 \times 10^{-4}\text{ s}^{-1}$. What fraction of the initial concentration of N_2O_5 will remain unreacted after exactly **1000 seconds** have passed? (Given: $10^{-0.217} = 0.606$). (2 Marks) [Chemical Kinetics] [PYQ 2014 / Practice] [Confidence: High]

Answer Key

Q1. (a) Shows positive deviation from Raoult's law. (A-B interactions are weaker than A-A and B-B, increasing vapour pressure and lowering the boiling point). **Q2.** (c) V^{3+} . (It has a $3d^2$ configuration with unpaired electrons, allowing d-d transitions. Sc^{3+} , Ti^{4+} , and Zn^{2+} have d^0 or d^{10} configurations). **Q3.** (c) The collision frequency of the reacting molecules. **Q4.** (b) The nitrogen atom of the NH_2 group farthest from the carbonyl group (hydrazine part). (The nitrogen attached to the carbonyl group has its lone pair involved in resonance, making it non-nucleophilic). **Q5.** (a) $\Delta_m^\circ(\text{NH}_4\text{Cl}) + \Delta_m^\circ(\text{NaOH}) - \Delta_m^\circ(\text{NaCl})$. **Q6.** (d) *Urea*. (Freezing point depression $\Delta T_f = i \cdot K_f \cdot m$. Urea is a non-electrolyte with $i = 1$, causing the least depression, hence the highest freezing point). **Q7.** (b) +3. **Q8.** (b) A hemiacetal. **Q9.** (b) It slowly oxidizes in air and sunlight to form the highly poisonous gas phosgene (COCl_2). **Q10.** (b) s^{-1} . (Overall order $n = 1/2 + 1/2 = 1$. First-order rate constant unit is s^{-1}). **Q11.** (c) 2-Methoxypropane. **Q12.** (d) Acetaldehyde (CH_3CHO). (It contains α -hydrogen atoms, so it undergoes Aldol condensation instead of Cannizzaro). **Q13.** (d) A is false but R is true. (Zinc is more reactive and will displace copper from the solution, destroying the vessel). **Q14.** (a) Both A and R are true and R is the correct explanation of A. **Q15.** (a) Both A and R are true and R is the correct explanation of A. **Q16.** (a) Both A and R are true and R is the correct explanation of A.

Q17. Ebullioscopic constant (K_b) is the elevation in boiling point produced when 1 mole of a non-volatile solute is dissolved in 1 kg of the solvent. Its SI unit is $K \text{ kg mol}^{-1}$. **Q18.** Direct current (DC) causes electrolysis of the solution, changing its concentration and composition near the electrodes. Alternating current (AC) prevents this electrolytic breakdown. **Q19.** A reaction that is truly of a higher order but behaves as a first-order reaction because one of the reactants is present in a massive excess. Example:

Acid-catalyzed hydrolysis of an ester. $\text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O} \xrightarrow{\text{H}^+} \text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH}$. (Water is in excess, Rate = $k'[\text{CH}_3\text{COOC}_2\text{H}_5]$). **Q20.** 1. Lanthanoids mainly show a +3 oxidation state (sometimes +2, +4), whereas Actinoids show a much wider range of oxidation states (+3, +4, +5, +6, +7). 2. Almost all Actinoids are radioactive, whereas most Lanthanoids (except Promethium) are non-radioactive.

Q21. (a) $K_3[\text{Al}(\text{C}_2\text{O}_4)_3]$. (b) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$.

Q22. Let molar mass of solute be M_2 . $W_2 = 30 \text{ g}$, $W_1 = 90 \text{ g}$. Moles of water $n_1 = 90/18 = 5 \text{ mol}$.

$$\frac{P^\circ - P}{P^\circ} = \frac{n_2}{n_1 + n_2} \approx \frac{n_2}{n_1}. \text{ Case 1: } \frac{P^\circ - 2.8}{P^\circ} = \frac{30/M_2}{5} = \frac{6}{M_2} \implies 1 - \frac{2.8}{P^\circ} = \frac{6}{M_2} \implies \frac{2.8}{P^\circ} = 1 - \frac{6}{M_2} \dots$$

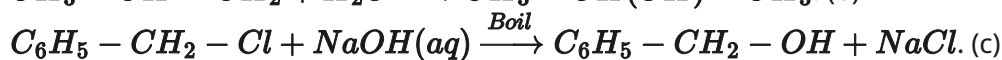
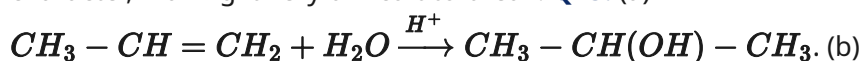
(Eq 1) Case 2: Added 18 g water. Total water = 108 g $\implies n'_1 = 6 \text{ mol}$.

$$\frac{P^\circ - 2.9}{P^\circ} = \frac{30/M_2}{6} = \frac{5}{M_2} \implies \frac{2.9}{P^\circ} = 1 - \frac{5}{M_2} \dots \text{ (Eq 2) Dividing (1) by (2):}$$

$$\frac{2.8}{2.9} = \frac{1 - 6/M_2}{1 - 5/M_2} \implies 2.8 - \frac{14}{M_2} = 2.9 - \frac{17.4}{M_2} \implies \frac{3.4}{M_2} = 0.1 \implies M_2 = 34 \text{ g mol}^{-1}. \text{ Q23. Cell}$$

constant $G^* = \text{Conductivity}(\kappa) \times \text{Resistance}(R)$.

$G^* = (0.146 \times 10^{-3} \text{ S cm}^{-1}) \times 1500 \Omega = 0.219 \text{ cm}^{-1}$. **Q24.** (a) To prevent its oxidation to poisonous phosgene gas (COCl_2) by air in the presence of sunlight. (b) The $\text{C} - \text{I}$ bond in iodoform is weak and easily breaks to give I^- ions which precipitate with AgNO_3 , whereas the $\text{C} - \text{Cl}$ bond in chloroform is much stronger. (c) Due to resonance, the $\text{C} - \text{X}$ bond in haloarenes acquires a partial double bond character, making it very difficult to break. **Q25.** (a)



(i) $\text{C}_6\text{H}_5 - \text{CH} = \text{N} - \text{NH} - \text{C}_6\text{H}_5$ (Benzaldehyde phenylhydrazone). (ii)



charged ions and vacant d-orbitals of appropriate energy to accept lone pairs of electrons from ligands. (b) The Mn^{2+} ion has a highly stable half-filled d^5 configuration. Removing an electron from it to form Mn^{3+} requires a massive amount of energy, making the reduction potential $E^\circ(\text{Mn}^{3+}/\text{Mn}^{2+})$ highly positive. (c) Zinc ($3d^{10}4s^2$) has completely filled d-orbitals in its ground state as well as in its common +2 oxidation state. **Q28.** Radioactive decay follows first-order kinetics. $k = \frac{0.693}{t_{1/2}} = \frac{0.693}{5730} \text{ year}^{-1}$.

$$t = \frac{2.303}{k} \log \left(\frac{[R]_0}{[R]} \right) = \frac{2.303 \times 5730}{0.693} \log \left(\frac{100}{80} \right) = \frac{2.303 \times 5730}{0.693} \log(1.25).$$

$$t = \frac{2.303 \times 5730 \times 0.0969}{0.693} \approx 1845 \text{ years}.$$

Q29. (a) Ionization isomerism. (b) Ethylenediamine (en) or Oxalate ($\text{C}_2\text{O}_4^{2-}$). (c) Draw square planar complexes. *cis*-isomer has Cl atoms at 90° , *trans*-isomer has Cl atoms opposite at 180° . The *trans*-isomer possesses a net zero dipole moment because the individual bond dipoles exactly cancel each other out.

Q30. (a) Fluorine is significantly more electronegative than Chlorine, creating a much stronger -I (electron-withdrawing) effect that stabilizes the resulting trifluoroacetate anion far more than the trichloroacetate anion. (b) It decreases the acidity of benzoic acid by intensifying the negative charge on the carboxylate anion, destabilizing it. (c) *p*-toluic acid < benzoic acid < *p*-nitrobenzoic acid. The $-\text{CH}_3$ group in *p*-toluic acid is an EDG (+I effect) which lowers acidity. The $-\text{NO}_2$ group is a strong EWG (-I, -R effects) which heavily stabilizes the conjugate base, increasing acidity.

Q31. (a) Kohlrausch's law states that the limiting molar conductivity of an electrolyte can be represented as the sum of the individual contributions of the anion and cation of the electrolyte. Application:

Calculation of limiting molar conductivity (Λ_m°) for weak electrolytes (e.g., acetic acid), which cannot be

obtained via direct extrapolation. (b) Oxidation (Anode): $H_2 \rightarrow 2H^+ + 2e^-$. Reduction (Cathode):

$Br_2 + 2e^- \rightarrow 2Br^-$. Overall: $H_2 + Br_2 \rightarrow 2H^+ + 2Br^-$. ($n = 2$).

$$E_{cell}^\circ = E_{cathode}^\circ - E_{anode}^\circ = 1.09 - 0.00 = +1.09 \text{ V}.$$

$$E_{cell} = E_{cell}^\circ - \frac{0.0591}{n} \log \left(\frac{[H^+]^2 [Br^-]^2}{P_{H_2}} \right) = 1.09 - \frac{0.0591}{2} \log \left(\frac{(0.030)^2 (0.010)^2}{1} \right).$$

$$Q = (9 \times 10^{-4}) \times (1 \times 10^{-4}) = 9 \times 10^{-8}. \log(9 \times 10^{-8}) = \log(9) - 8 = 0.954 - 8 = -7.046.$$

$$E_{cell} = 1.09 - 0.02955 \times (-7.046) = 1.09 + 0.208 = 1.298 \text{ V}.$$

Q32. (a) Compound A is Phenol (C_6H_5OH). Compound B is Sodium salicylate ($C_6H_4(OH)COONa$). Compound C is Salicylic acid / 2-Hydroxybenzoic acid ($C_6H_4(OH)COOH$). Compound D is Aspirin / Acetylsalicylic acid ($C_6H_4(OCOCH_3)COOH$).

(b) Kolbe's Reaction. **Q33.** (a) For a first order reaction $R \rightarrow P$,

$$Rate = -\frac{d[R]}{dt} = k[R]. \frac{d[R]}{[R]} = -kdt. \text{ Integrating both sides from } t = 0 \text{ to } t = t:$$

$$\int_{[R]_0}^{[R]} \frac{d[R]}{[R]} = -k \int_0^t dt \Rightarrow \ln[R] - \ln[R]_0 = -kt \Rightarrow \ln \left(\frac{[R]}{[R]_0} \right) = -kt \Rightarrow k = \frac{2.303}{t} \log \left(\frac{[R]_0}{[R]} \right)$$

$$. (b) k = \frac{2.303}{t} \log \left(\frac{[R]_0}{[R]} \right) \Rightarrow \log \left(\frac{[R]_0}{[R]} \right) = \frac{kt}{2.303} = \frac{5.0 \times 10^{-4} \times 1000}{2.303} = \frac{0.5}{2.303} = 0.217.$$

$$\frac{[R]_0}{[R]} = \text{antilog}(0.217) = 10^{0.217}. \text{ Fraction remaining} = \frac{[R]}{[R]_0} = 10^{-0.217} = 0.606. \text{ Thus, } 60.6\% \text{ remains unreacted.}$$

CBSE Class 12 (2027) — Half-Yearly Predicted Paper (80% syllabus). AI-generated via PYQ/Blueprint analysis, not actual exam questions.